

1.To-Do List Application:

Create a Python program that allows users to add, view, mark as complete, and delete tasks from a to-do list.

```
tasks = []

def add_task(task):

    tasks.append(task)

def view_tasks():

    for task in tasks:

        print(f"- {task}")

def delete_task(task):

    tasks.remove(task)

add_task("Complete Python project")

add_task("Read a book")

view_tasks()

delete_task("Read a book")

view_tasks()
```

Output:

- Complete Python project
- Read a book
- Complete Python project

2.Simple Contact Book:

Write a Python program that allows users to add, view, update, and delete contact information (name, phone number, and email).

```
contacts = {}
```

```
def add_contact(name, phone, email):  
    contacts[name] = {"Phone": phone, "Email": email}  
  
def view_contact(name):  
    if name in contacts:  
        print(f"Name: {name}, Phone: {contacts[name]['Phone']}, Email:  
{contacts[name]['Email']}")  
  
def delete_contact(name):  
    if name in contacts:  
        del contacts[name]  
  
add_contact("John", "555-1234", "john@example.com")  
  
view_contact("John")  
  
delete_contact("John")
```

Output:

Name: John, Phone: 555-1234, Email: john@example.com

3.Simple ATM System:

Develop a Python program simulating an ATM system where users can check their balance, deposit money, and withdraw money.

```
balance = 1000
```

```
def check_balance():  
    print(f"Your current balance is ${balance}")  
  
def deposit(amount):  
    global balance  
    balance += amount  
  
def withdraw(amount):
```

```
global balance
```

```
if amount <= balance:
```

```
    balance -= amount
```

```
check_balance()
```

```
deposit(200)
```

```
withdraw(500)
```

```
check_balance()
```

Output:

Your current balance is \$1000

Your current balance is \$1200

Your current balance is \$700

4.Temperature Converter:

Write a Python program that converts temperatures between Celsius and Fahrenheit.

```
def celsius_to_fahrenheit(celsius):
```

```
    return (celsius * 9/5) + 32
```

```
def fahrenheit_to_celsius(fahrenheit):
```

```
    return (fahrenheit - 32) * 5/9
```

```
celsius = 25
```

```
fahrenheit = celsius_to_fahrenheit(celsius)
```

```
print(f"{celsius}°C is {fahrenheit}°F")
```

Output:

25°C is 77.0°F

5.Simple Inventory System:

Create a Python program to manage an inventory system, allowing users to add, update, view, and delete products.

```
inventory = {}

def add_product(name, quantity):

    inventory[name] = quantity

def update_product(name, quantity):

    if name in inventory:

        inventory[name] = quantity

def view_inventory():

    for name, quantity in inventory.items():

        print(f"Product: {name}, Quantity: {quantity}")

add_product("Apples", 100)

update_product("Apples", 120)

view_inventory()
```

Output:

Product: Apples, Quantity: 120

6.Basic Email Validation:

Write a Python program that validates email addresses using regular expressions to ensure they follow the correct format.

```
import re

def validate_email(email):

    regex = r'^[a-zA-Z0-9_+-.]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-]+$'

    if re.match(regex, email):
```

```
        print("Valid email address!")

    else:

        print("Invalid email address!")

validate_email("john.doe@example.com")

validate_email("john.doe@com")
```

Output:

Valid email address!

Invalid email address!

7.Currency Converter:

Develop a Python program that converts an amount in one currency to another, given the exchange rate.

```
def convert_currency(amount, exchange_rate):

    return amount * exchange_rate

usd_amount = 100

exchange_rate = 0.85

eur_amount = convert_currency(usd_amount, exchange_rate)

print(f"${usd_amount} is equivalent to €{eur_amount}")
```

Output:

\$100 is equivalent to €85.0

8.Simple Alarm Clock:

Write a Python program that sets an alarm for a specific time and notifies the user when the alarm time is reached.

```
import time

def set_alarm(hour, minute):
```

```
current_time = time.localtime()

alarm_time = time.struct_time((current_time.tm_year, current_time.tm_mon,
current_time.tm_mday, hour, minute, 0, 0, 0, 0))

while True:

    if time.localtime() >= alarm_time:

        print("ALARM! Time's up!")

        break

    time.sleep(60)

set_alarm(12, 30)
```

Output:

ALARM! Time's up!

9.Basic Voting System:

Create a Python program that allows users to vote for their favorite candidate and displays the total votes for each candidate.

```
votes = {"Alice": 0, "Bob": 0, "Charlie": 0}

def vote(candidate):

    if candidate in votes:

        votes[candidate] += 1

def show_results():

    for candidate, vote_count in votes.items():

        print(f"{candidate}: {vote_count} votes")

vote("Alice")

vote("Bob")

vote("Alice")
```

```
show_results()
```

Output:

Alice: 2 votes

Bob: 1 vote

Charlie: 0 votes

10.Simple Loan Calculator:

Write a Python program that calculates the monthly installment for a loan based on the loan amount, interest rate, and loan period.

```
def calculate_loan_payment(principal, rate, time):  
    rate = rate / (12 * 100)  
    time = time * 12  
    payment = (principal * rate) / (1 - (1 + rate) ** -time)  
    return payment  
  
loan_amount = 5000  
interest_rate = 5  
loan_period = 2  
  
monthly_payment = calculate_loan_payment(loan_amount, interest_rate,  
loan_period)  
  
print(f"Your monthly payment is ${monthly_payment:.2f}")
```

Output:

Your monthly payment is \$219.36